

HUBERT KIM

AUTOMATION RESEARCH ENGINEER

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FAVORITE TOOLS

Simulation/Optimization Tools

- Python | pyTorch
- Python | pyDrake
- Python | pyBullet
- Python | casadi
- MATLAB

Electronics

- Click PLC
- TI ARM Processor

Implementation Platform

- ABB Robots
- Fanuc Robots

Sensors, Hardware

- Basler 2D Camera
- Keyence profilometer
- Zivid

CAD Design

- Solidworks
- NX

EDUCATION

PhD,

Mechanical Engineering
Virginia Tech, Blacksburg, VA
Earned in Dec 2021

BS, *cum laude*,

Mechanical Engineering
NYU Tandon, Brooklyn, NY
Earned in May 2015

SUMMARY

Hardware-based roboticist practicing short-horizon motion tasks. Competencies:

- Delivering motion planning for Industrial robots for a collaborative task.
- Practicing succinct theoretical tools based on the physical phenomena.
- Developing methods for low-TRL (< 5) and approaching with short-term (<1 year) and long-term (>3 years) solutions.

Trajectory Optimization | Collaborative industrial robots | Robot Perception

AREAS OF EXPERTISE

AUTOMATION

Automation Research Engineer | **GE Aerospace Research Center**

- Established a multi-robot cell with an HMI platform via TCP/IP, providing three-robot coordinated motion with offline trajectory planning feature.
- Improved the off-the-shelf motion function (linear and SLERP interpolation) configuration and reachability by offline programming method, improving the process time efficiency by ~30%.
- Improved the precision by controlling error boundary control (GR&R) on the optical system (2.5 mil) to increase the ABB robot's factory resolution during the pick and place task on the 3D contour surface

MECHATRONICS

Mechatronics Research Engineer | **GE Aerospace Research Center**

- Prototyped a surveillance robot to solve the accessibility problem under a confined space while keeping its structural rigidity (IP filed & Patent Award)
- Designed a miniaturized EOAT for compliant material handling

Research Assistant | *Assistive Robotics Laboratory at Virginia Tech*

- Developed a custom direct-drive BLDC actuator to meet the critical quality (weight- 500 g, cost- \$ 509) while providing registry-level customizability in data communication as exhibited in [HardwareX](#)

Signal Processing and Analysis

Undergraduate Researcher | *Dynamic System Laboratory at NYU*

- Performed impedance analysis on the smart materials to characterize the material's electro/mechanical properties using a potentiostat, ending with a journal article from [J. of Intell Mater Syst Struct](#)
- Developed and implemented the energy harvesting method using the signal processing skills (system identification and impedance matching), and ended up improving the power delivery by more than 60 %—[Smart Materials and Structures](#)
- Leading to winning a Best Mechanical Engineering Experience Award for Undergraduate from NYU, and ICTAS funding for doctoral study at Virginia Tech

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AUTOMATION RESEARCHER ENGINEER

SOFT SKILL SET

Deriving long & short-term approaches via RCA

Controlling Error boundary via GR&R

Quick Proof of Concept with Most Viable Prototype

Improving the operational safety margin

ACHIEVEMENTS

Spotlight Award
GE Aerospace Research
April 2025

Patent Award
GE Aerospace Research
Jan 2025

Applause Award
GE Aerospace Research
August 2024

Bravo Award
GE Aerospace Research
Oct 2023

Doctoral Scholarship
ICTAS, Virginia Tech
March 2016–July 2020

Best Mechanical Engineering Experience Award for Undergraduate
NYU Tandon
May 2015

PROFESSIONAL EXPERIENCE

AUTOMATION/MECHATRONICS ENGINEER Aug 2022 – Current
Mechanical Components and Systems Lab | GE Aerospace Research Center

Automated manufacturing system to productivity cost saving (expect to reduce 6 Direct Labor Hours) and improvement of part assembly quality & yield. Key achievements include:

- **Spotlight award** for reaching a significant 1st milestone on the project by delivering and demonstrating the high-quality automation result on time.
- **Applause award** for supporting the offsite leadership tour.
- **Bravo award** for demonstrating real grit in the innovation of robotic manufacturing. This demonstration showed the feasibility to the Aerospace senior leadership and was recognized as an essential technology for the program.

GRADUATE RESEARCH ASSISTANT May 2015 – Dec 2021
Assistive Robotics Laboratory | Virginia Tech Advisor: Dr. Alan T. Asbeck

The physiological states of the wearer from the exoskeleton and the robot's perception were studied to understand the user's intention.

Dissertation: "Joint Torque Feedback for Arm Motion Training"

UNDERGRADUATE RESEARCH ASSISTANT May 2013 – Dec 2015
Dynamic System Laboratory | NYU Advisor: Dr. Maurizio Porfiri

System identification: understand and apply frequency domain modeling theory to real-world applications.

Thesis: "Voltage attenuation along the electrodes of ionic polymer metal composites"

SELECTED PUBLICATIONS

The Effects of Torque Magnitude and Stiffness in Arm Guidance through Joint Torque Feedback

Kim, H., Asbeck, A.

- 2022
- [IEEE Access](#)

Just Noticeable Differences for Elbow Joint Torque Feedback

Kim, H., Asbeck, A.

- 2021
- [Scientific Report](#)

An Elbow Exoskeleton for Haptic Feedback Made with a Direct Drive Hobby Motor

Kim, H., Asbeck, A.

- 2020
- [HardwareX](#)